

WHAT DOES A MODEL USE TO PREDICT?

↑
INFORMATIVE MECHANISM

DATA-DRIVEN MODELING →

FIRST PRINCIPLES

PARTICLE PHYSICS

ORBITAL DYNAMICS

DIRECT NUMERICAL
SIMULATION (FLUIDS)

SEISMIC

TOMOGRAPHY

GEODYNAMO

MULTISCALE
MECHANICS

NUMERICAL
WEATHER
PREDICTION

CLIMATE
PREDICTION

MOLECULAR
DYNAMICS

NATURAL LANGUAGE
PROCESSING

CLASSIFYING CATS

ALPHA ZERO

TOO HARD
NO SKILL

→
INFORMATIVE DATA

RELIABLE
DATA & MECHANISM

↑
INDUCTIVE BIASES
(BETTER MECHANISM)